P FOR REPRESENTABLE MORPHISMS IS EQUIVALENT TO P ON ALL ALGEBRAIC-SPACE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 649-659 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 661-675, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 34 of queue_stacks-properties.json (pb=486, stmt 649-659, proof 661-675).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let P be a property of morphisms of algebraic spaces as above. Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks representable by algebraic spaces. The following are equivalent:*

(1) *f has P,*
(2) *for every algebraic space Z and morphism $Z \to \mathcal{Y}$ the morphism $Z \times_{\mathcal{Y}} \mathcal{X} \to Z$ has P.*

Proof. The implication (2) $\Rightarrow$ (1) is immediate. Assume (1). Let $Z \to \mathcal{Y}$ be as in (2). Choose a scheme U and a surjective étale morphism $U \to Z$. By assumption the morphism $U \times_{\mathcal{Y}} \mathcal{X} \to U$ has P. But the diagram

$$\begin{array}{ccc} U \times_{\mathcal{Y}} \mathcal{X} & \longrightarrow & Z \times_{\mathcal{Y}} \mathcal{X} \\ \downarrow & & \downarrow \\ U & \longrightarrow & Z \end{array}$$

is cartesian, hence the right vertical arrow has P as $\{U \to Z\}$ is an fppf covering. □

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.